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BUILDING A FILE SHARING APPLICATION

Network Programming - IT4062E - 1st Progress Report

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ONE LOVE. ONE FUTURE.

1. Actors & Problem Definition

System Actors

- **Regular Member**:
  - Registers & signs in to the system.
  - Requests to join groups.
  - Uploads/downloads files, creates subfolders in group space.
- **Group Admin** (Group Creator):
  - Creates and manages sharing groups.
  - Exclusive rights: rename & delete files and subfolders.

Problem Definition

- **Group-Centric Collaboration**: Provide dedicated shared folders for teams to organize and collaborate online.
- **Hierarchical Content Organization**: Allow members to structure project data with nested subfolders and file uploads.
- **Role-Based Access Control**: Enforce administrative authority so that only group admins can rename or delete files and folders, preventing unauthorized data loss.
- **Network File Transfer**: Support secure client-server file uploads and downloads across the network.

2. Core Use Cases

General Use Cases

- **UC1 — Register & Log In**
  Create an account and authenticate for usage.
- **UC2 — Create or Join Group**
  Create a new group or request to join existing ones.
- **UC3 — Upload & Download Files**
  Upload files to group storage and download shared files.
- **UC4 — Create Subfolders**
  Create subdirectories within the group folder.

Admin-Only Use Cases

- **UC5 — Delete Files & Subfolders**
  Exclusive authority to remove files and nested folders within the group space.
- **UC6 — Rename Files & Folders**
  Exclusive authority to rename files and directories.
- **UC7 — Manage Group Membership**
  Review, accept, or reject user requests to join the group.

3. Technology Stack & Key Challenges

Technology Stack

- **Core Language: Java (JDK 17+)**
  Object-oriented design, cross-platform portability, robust standard I/O.
- **Transport Protocol: TCP Sockets**
  Connection-oriented `java.net` sockets ensuring reliable, lossless byte streams.
- **Concurrency: Multithreaded Engine**
  Thread pools (`ExecutorService`) to handle concurrent client sessions.

Challenges

- **Concurrent Access**
  Handling conflicts when an admin deletes or renames a folder while members are uploading or downloading inside it.
- **Partial Transfers**
  Detecting broken TCP pipes mid-stream; rolling back incomplete uploads to prevent corrupted files on the server.
- **Authentication Across Requests**
  Maintaining session state and validating identity/tokens across subsequent commands without re-sending credentials.



THANK YOU !